

DD2380 Artificial Intelligence

Introduction



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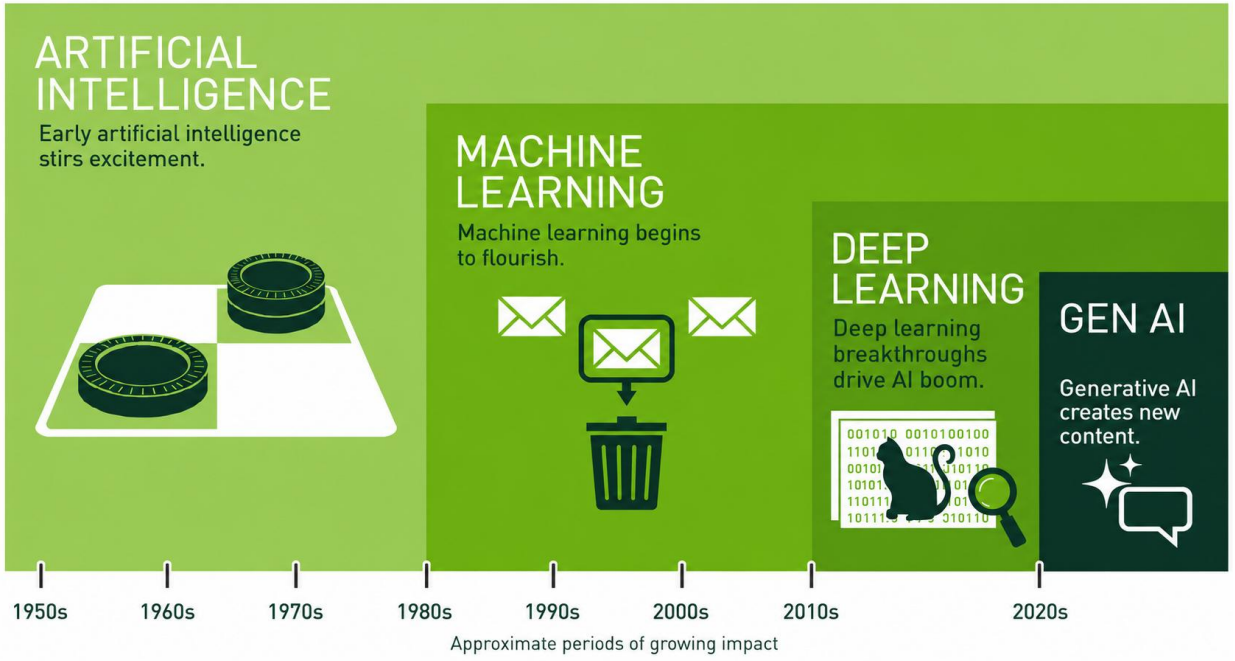
Mentimeter 1

What is Artificial Intelligence?

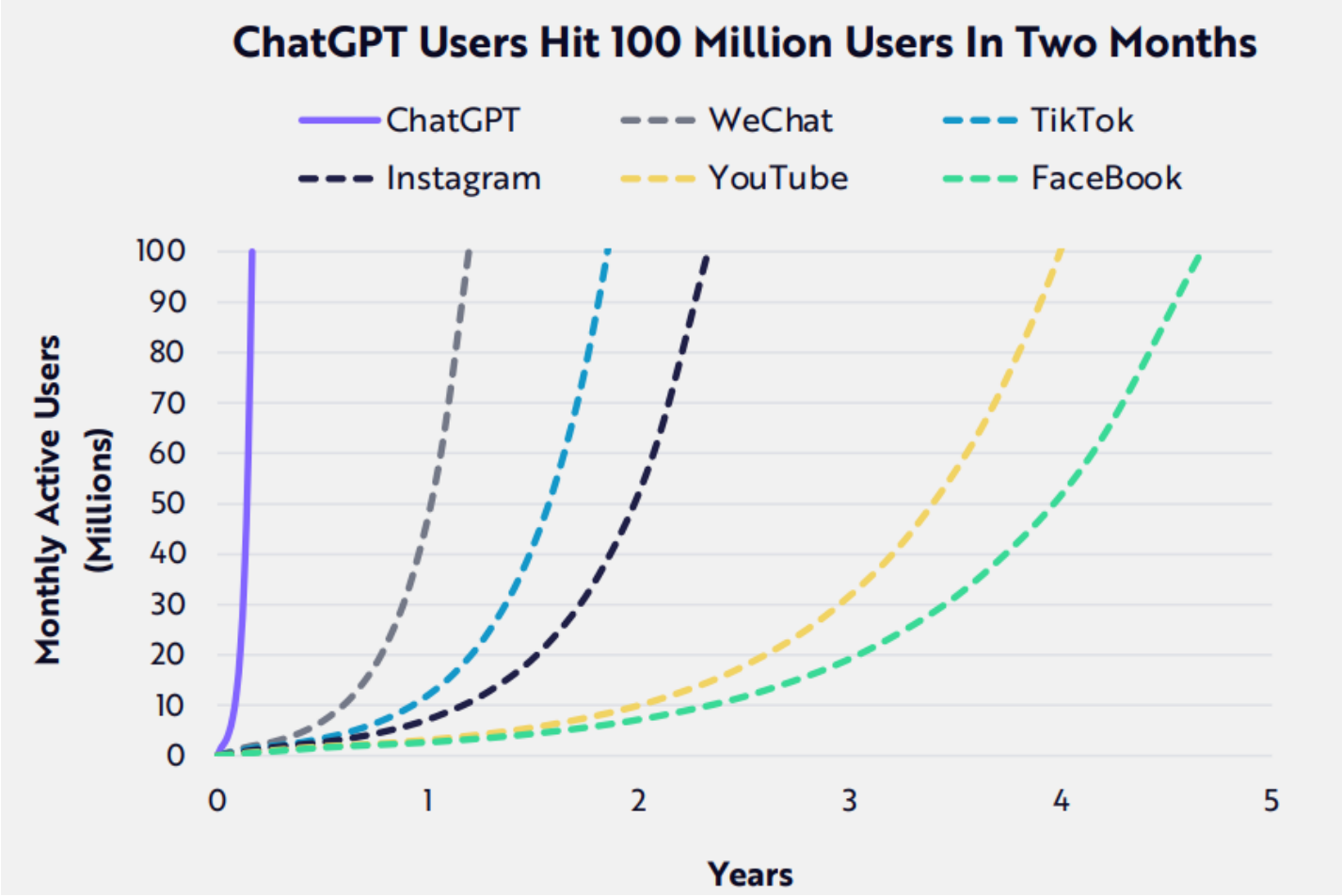
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AI is not a new thing

- 1956: Dartmouth Conference, AI term is born.
- 1959: computers play checkers better than the average human
- 1974-80: funding cuts (first “AI winter”).
- 1985: billion dollars spent on expert systems
- 1987: market for specialized AI hardware collapses
- 1997: IBM’s Deep Blue beats Garry Kasparov
- 2012: Deep learning with ImageNet breakthrough
- 2016: AlphaGo beats Lee Sedol
- 2017: Attention is all you need (Transformers)
- 2022: ChatGPT
- …



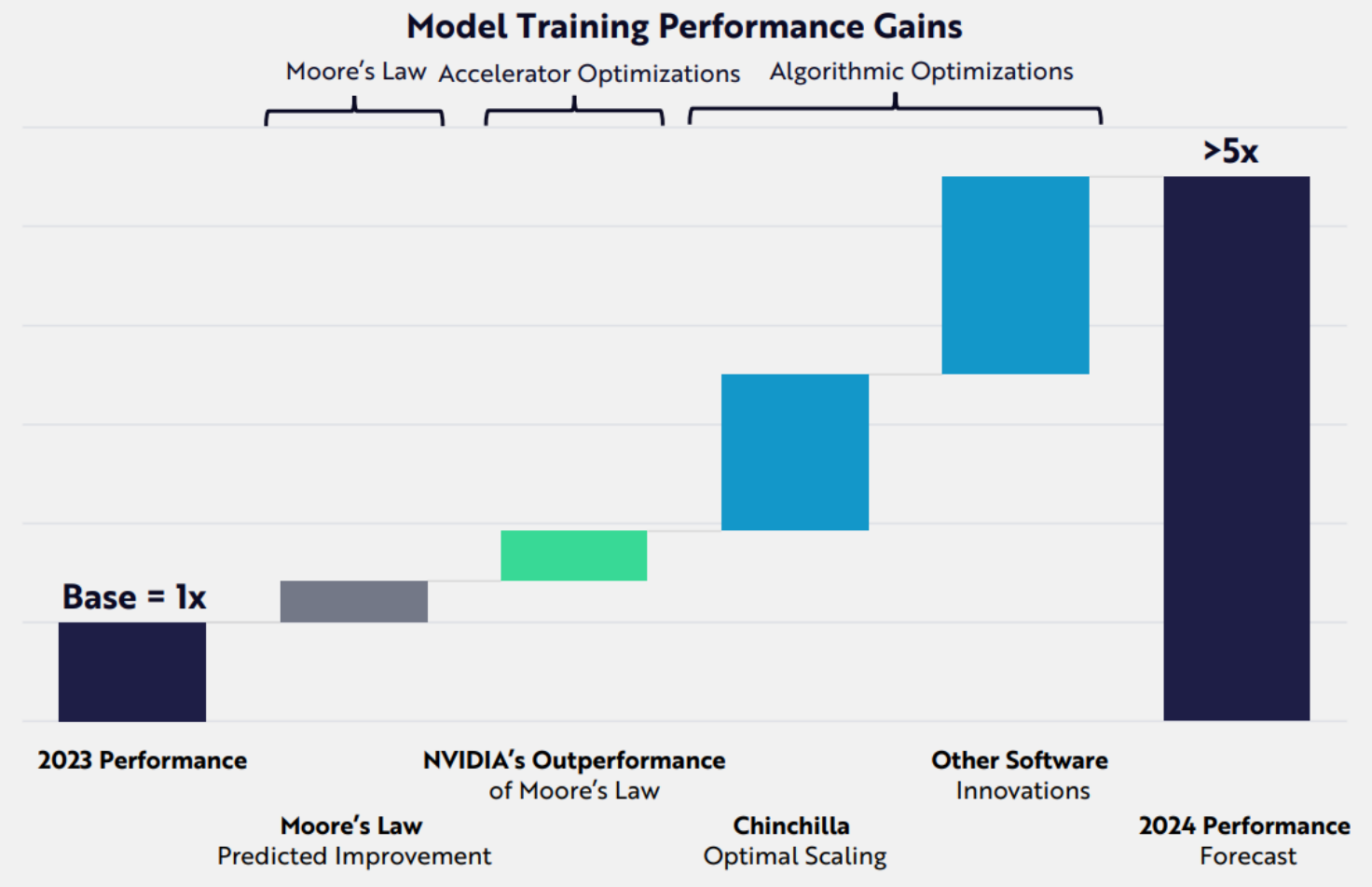
ChatGPT boom



Between June 2022 and March 2023, global searches for 'AI' almost quadrupled from 7.9 million to 30.4 million

ChatGPT user update:
November 2023: ~100 M
August 2026 > 1.000 M

Rate of growth of training performance

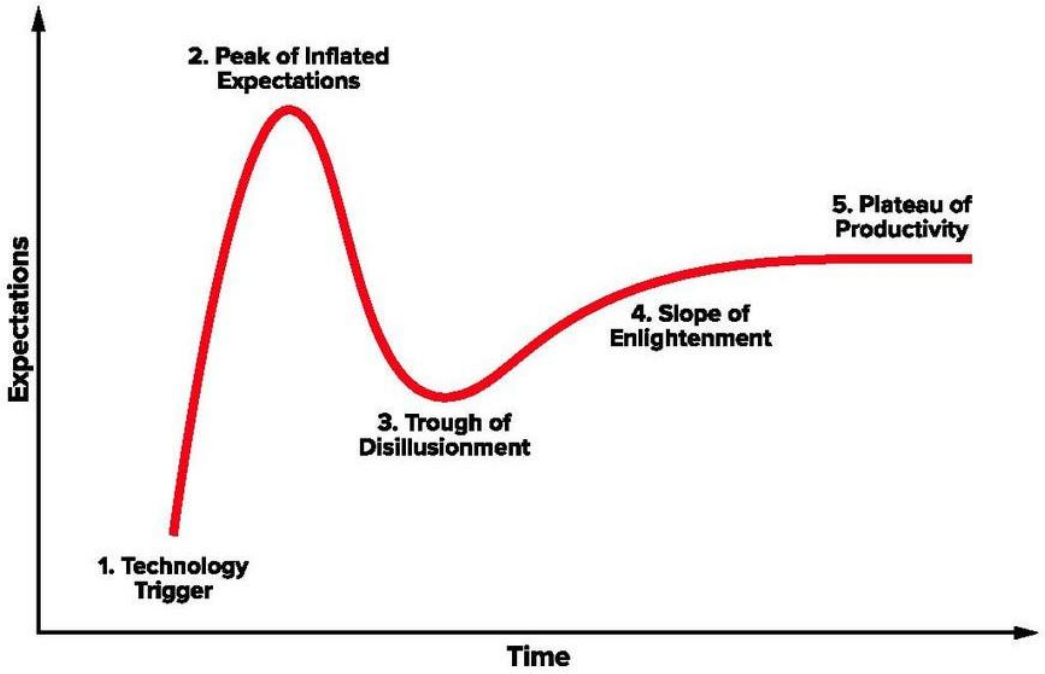


Sources: ARK Investment Management LLC, 2024. This ARK analysis is based on a range of data sources, including Benaich 2023, Touvron et al. 2023, Yang et al. 2023, Leviathan et al. 2022, and Dao 2023

Gartner’s Hype Cycle

- A model that describes the typical progression of new technologies from initial excitement to mainstream adoption. The hype cycle has five stages:

1. **Innovation Trigger** – A new technology emerges and garners interest.
2. **Peak of Inflated Expectations** – Hype builds as people start seeing potential applications, often leading to overblown expectations.
3. **Trough of Disillusionment** – The initial excitement fades, and limitations or challenges become more apparent.
4. **Slope of Enlightenment** – Some companies and users continue developing and understanding the technology's practical uses and realistic applications.
5. **Plateau of Productivity** – The technology matures, stabilizes, and finds its niche in the market, leading to mainstream adoption and steady progress.





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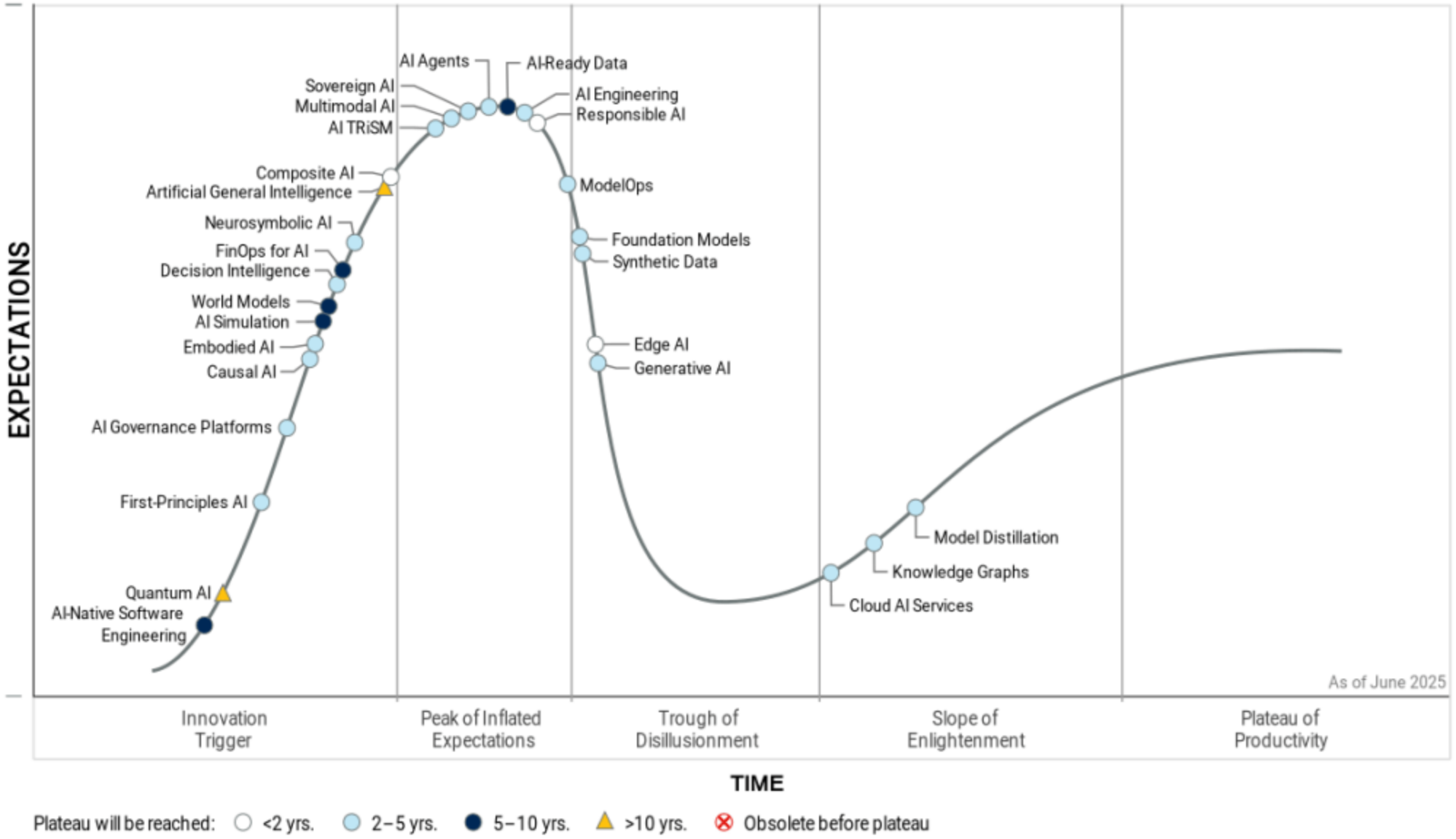


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Mentimeter 2

Where is **generative AI** in this year's hype cycle?

Hype Cycle for Artificial Intelligence, 2025



Gartner





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Mentimeter 3

I use LLMs to help me in my studies



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Harvard Undergraduate Study

- A survey of 326 Harvard students (August 2024) revealed that 87.5 percent use generative AI, at least weekly.
- ChatGPT is by far the most popular. AI is often used to answer questions, as a replacement for traditional sources of information such as Wikipedia and to support writing and programming.

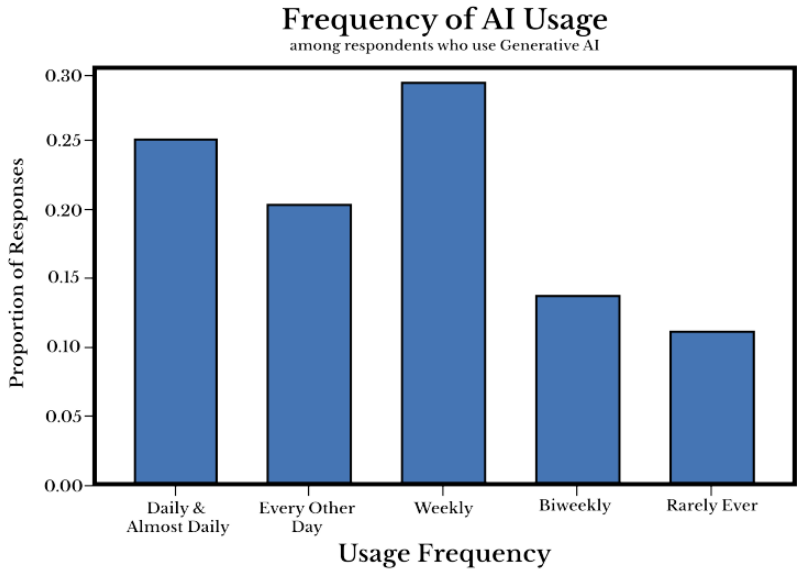
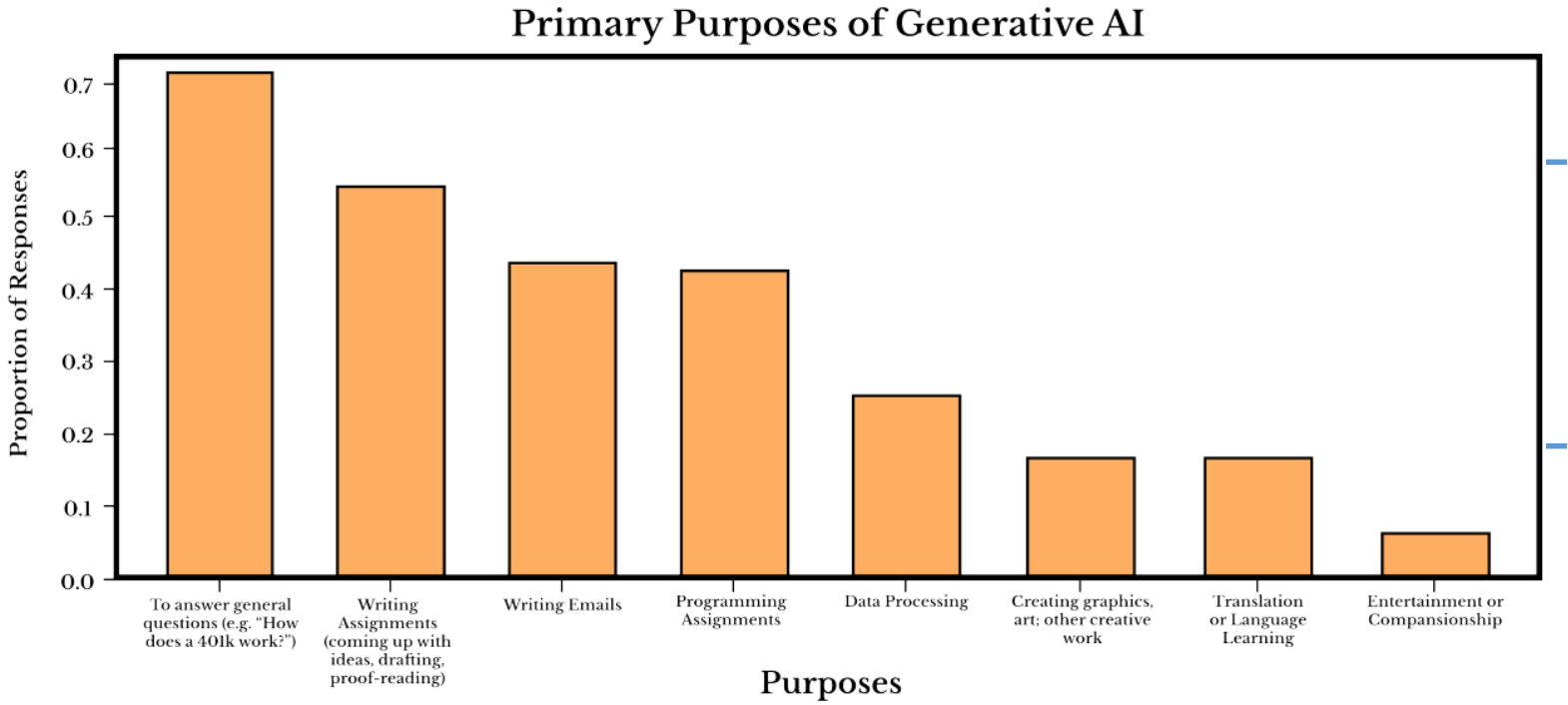


Figure 1: Most Harvard students who use generative AI report using it at least once a week.







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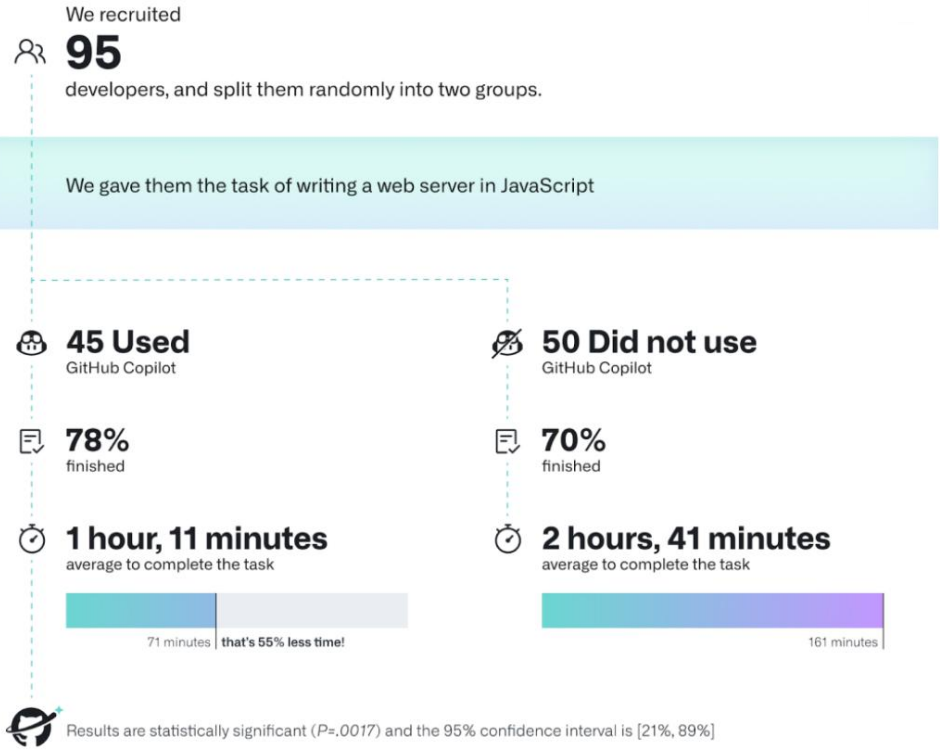
Mentimeter 4

I use LLMs when programming



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Programming with Github Copilot



Source: quantifying GitHub Copilot's impact on developer productivity and happiness - The GitHub Blog (2022)



Amazon

- **Amazon Web Services** CEO, **Matt Garman**, reportedly hinted at the possibility of artificial intelligence taking over a significant portion of coding tasks.
- “If you go forward 24 months from now, or some amount of time — I can’t exactly predict where it is — it’s possible that most developers are not coding,” said Garman



Andy Jassy • 3rd+
President and CEO at Amazon
1w •

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One of the most tedious (but critical tasks) for software development teams is updating foundational software. It’s not new feature work, and it doesn’t feel like you’re moving the experience forward. As a result, this work is either dreaded or put off for more exciting work—or both.

Amazon Q, our GenAI assistant for software development, is trying to bring some light to this heaviness. We have a new code transformation capability, and here’s what we found when we integrated it into our internal systems and applied it to our needed Java upgrades:

- The average time to upgrade an application to Java 17 plummeted from what’s typically 50 developer-days to just a few hours. We estimate this has saved us the equivalent of 4,500 developer-years of work (yes, that number is crazy but, real).

- In under six months, we've been able to upgrade more than 50% of our production Java systems to modernized Java versions at a fraction of the usual time and effort. And, our developers shipped 79% of the auto-generated code reviews without any additional changes.

- The benefits go beyond how much effort we’ve saved developers. The upgrades have enhanced security and reduced infrastructure costs, providing an estimated $260M in annualized efficiency gains.

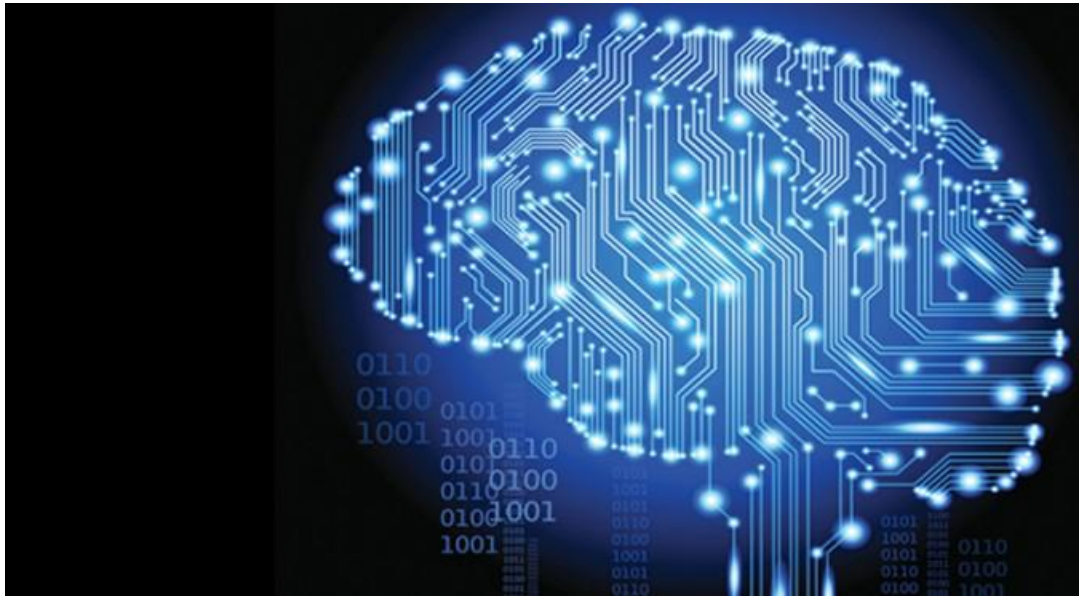
This is a great example of how large-scale enterprises can gain significant efficiencies in foundational software hygiene work by leveraging Amazon Q. It’s been a game changer for us, and not only do our Amazon teams plan to use this transformation capability more, but our Q team plans to add more transformations for developers to leverage.

Narrow (weak) AI vs. general (strong) AI

Solves a single task, even better than a human



Solves any task a human can solve without being tailored for it

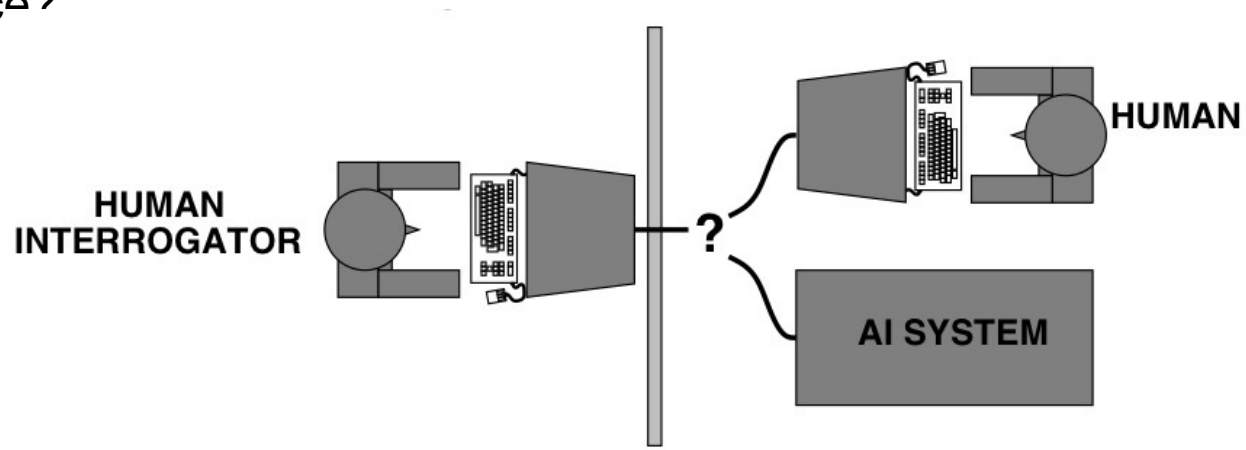


General (strong, full) AI

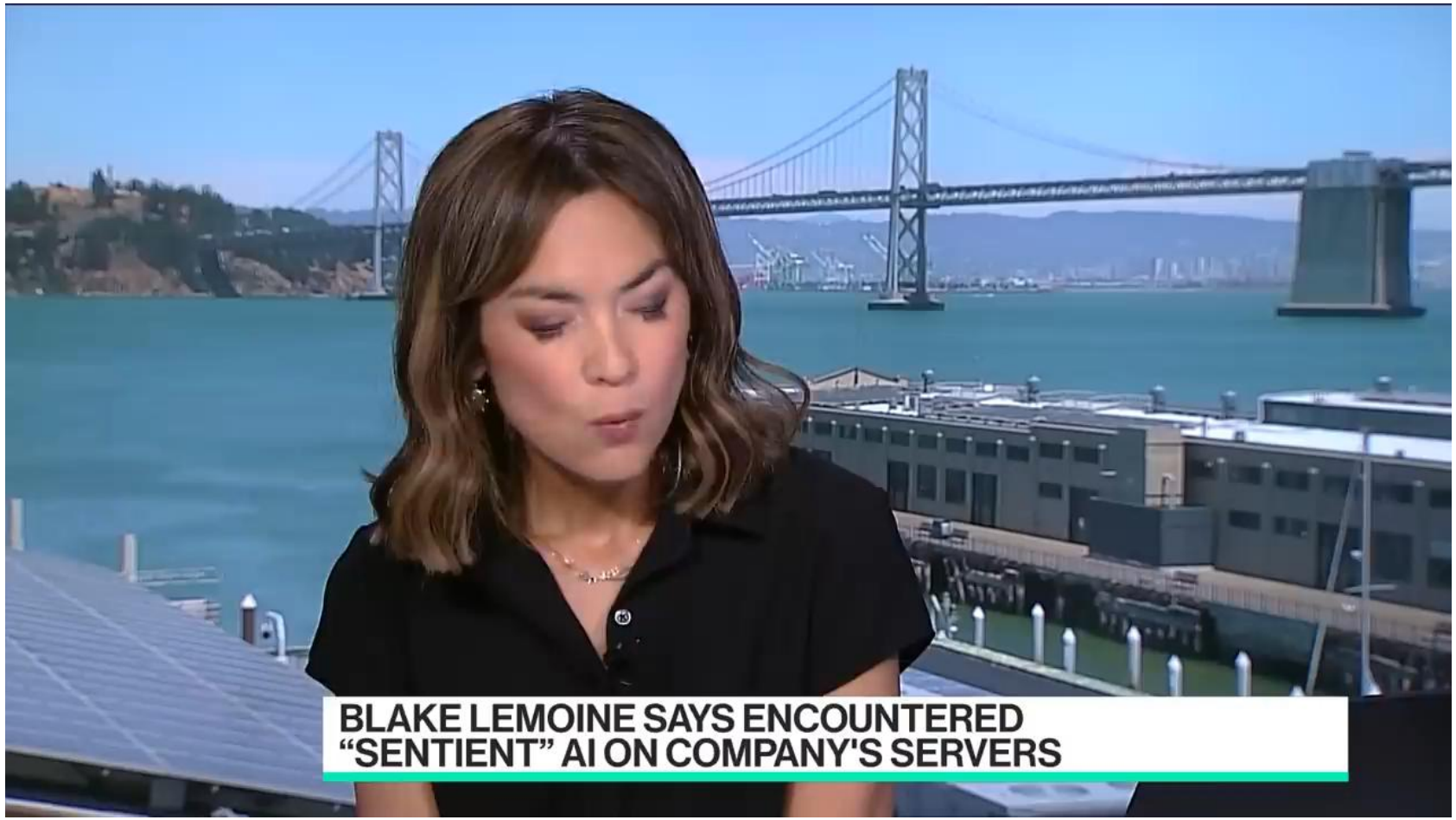
A system that solves any task a human can solve without being tailored for it

How could we test for general intelligence?

- ”IKEA” Test
- The Robot College Student Test
- The Employment test
- The Coffee Test
- The Turing Test



2022: Blake Lemoine on LaMDA





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Mentimeter 5

When will we achieve Artificial General Intelligence (AGI)?



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AI “today”

Andrew Ng in January 25, 2017, feels even more convincing today:

- “Just as electricity transformed almost everything 100 years ago, today I actually have a hard time thinking of an industry that I don’t think AI will transform in the next several years,”

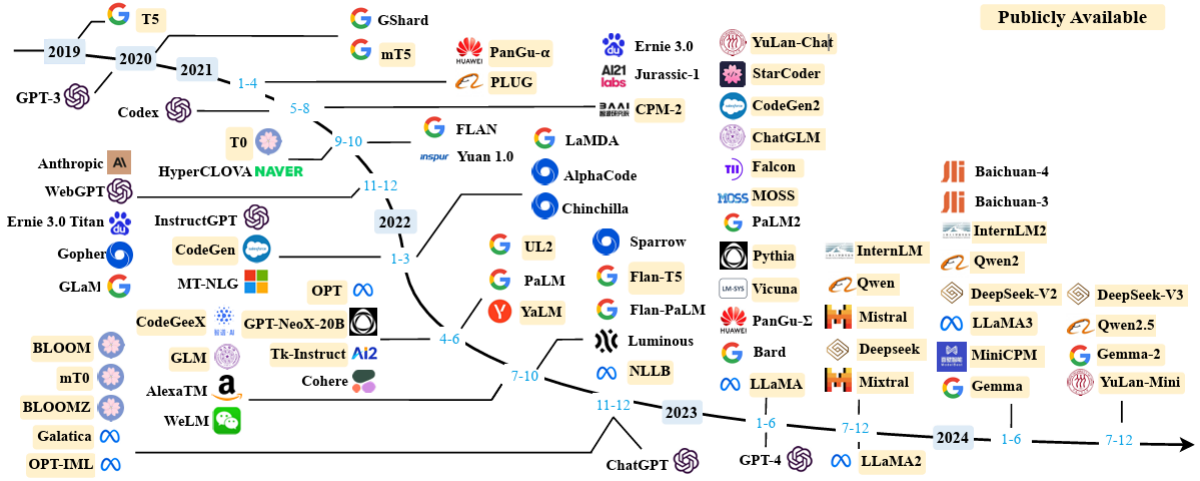


Where we are today: NLP and Gen AI

(2023

A Survey of Large Language Models

Wayne Xin Zhao, Kun Zhou*, Junyi Li*, Tianyi Tang, Xiaolei Wang, Yupeng Hou, Yingqian Min, Beichen Zhang, Junjie Zhang, Zican Dong, Yifan Du, Chen Yang, Yushuo Chen, Zhipeng Chen, Jinhao Jiang, Ruiyang Ren, Yifan Li, Xinyu Tang, Zikang Liu, Peiyu Liu, Jian-Yun Nie and Ji-Rong Wen



ChatGPT for Good? On Opportunities and Challenges of Large Language Models for Education

Enke… Dem… Hülle… Pfeff… Stadl…

Abs… Larg… to fu… mod… appl… state… educ… rega… sets of compet… unexpected bri… pedagogical ap… integrate and t… challenges suc… for misuse are… challenges can… societal biases… these challeng…

Keywords
Large languag…

[1] Technical Universi…
[2] Ludwig-Maximilian…
[3] University of Tübin…
*Corresponding au…

Scientists' Perspectives on the Potential for Generative AI in their Fields

Programming Is Hard – Or at Least It Used to Be: Educational Opportunities and Challenges of AI Code Generation

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Figure 1: An image generated by Midjourney with the prompt "robot writing computer code while student watches, computer screens, computer programming, computer code, realistic, highly detailed, cinematic –aspect 16:9"

Where we are today: image understanding





Where we are today: autonomous driving



Navlab 1 autonomous vehicle (86-90, CMU)

Waymo Fully-Autonomous Vehicles

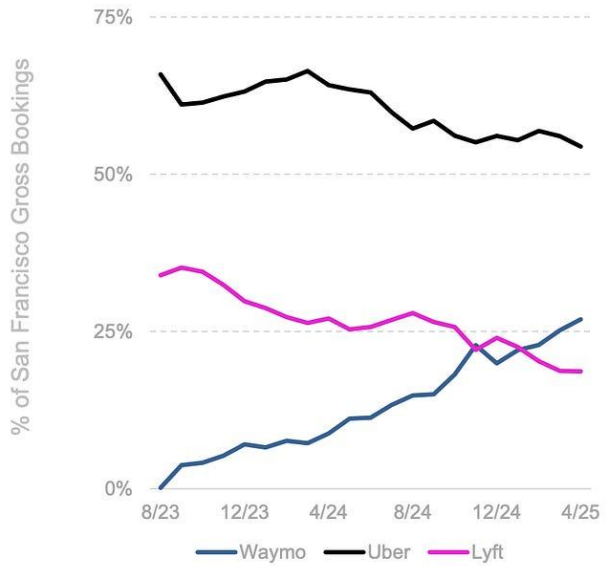
[We are creating] an end-to-end, very, very robust, and large end-to-end system that's multi-modal in its foundation so that perception planning and prediction… can become even more robust than it is today.

- Waymo Co-CEO Tekedra Mawakana, 1/25

What we've done in San Francisco is prove to ourselves – and to the world – that not only does autonomy work, but it works at scale in a market and can be a viable commercial product.

- Waymo Co-CEO Dmitri Dolgov, 3/25

Estimated Market Share (Gross Bookings) – 8/23-4/25, San Francisco Operating Zone, per YipitData



Where are we today: robotics





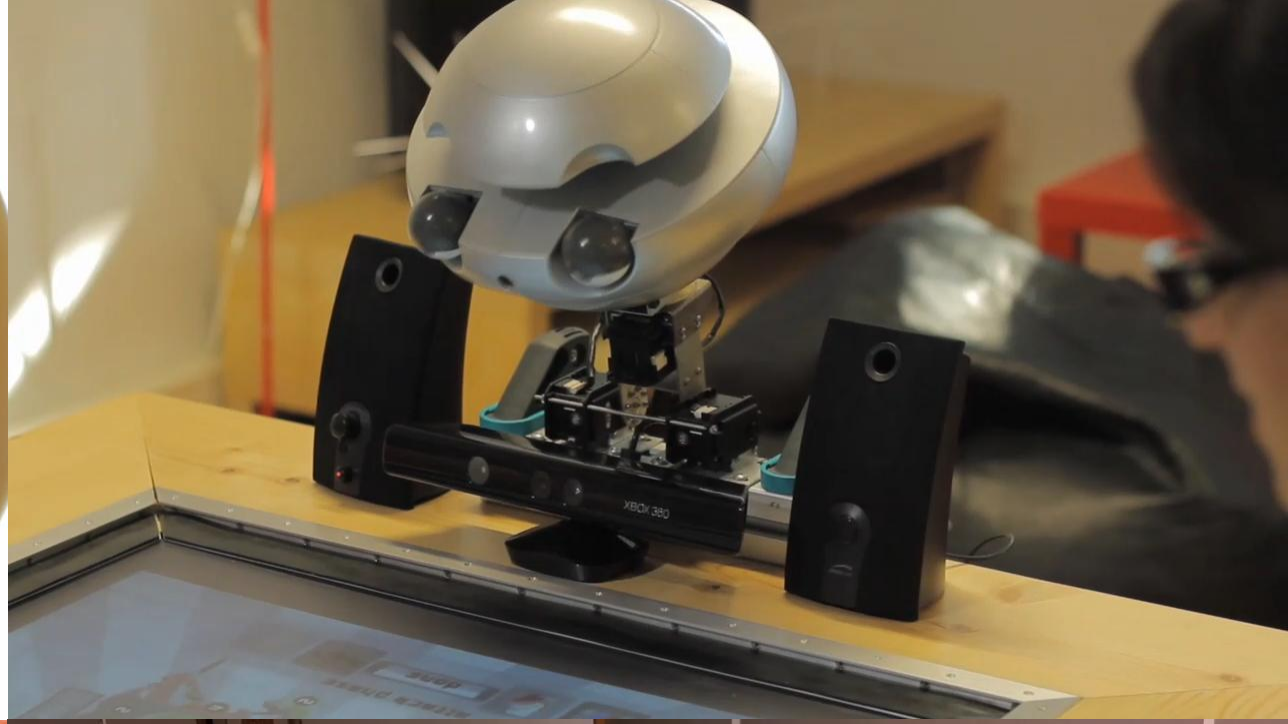

Unitree G1
Mass Production Version
Leap into the Future!



About me

- Andre Pereira
  - Associate Professor in Human-Robot Interaction
  - Joined KTH in 2018
- Previously
  - Masters and PhD (2014) – IST Technical University of Lisbon
  - Postdoc – Yale University
  - Senior Research Associate – Disney Research Pittsburgh
  - Senior Solutions Architect – Furhat Robotics
- Teaching
  - Course responsible for DD2380 – Artificial Intelligence
  - Course responsible for FDT3304 – Topics in Social Robotics
  - Teacher, assistant for DD2421 – Machine Learning
  - Assistant for DT2140 – Multimodal Interaction and Interfaces




XBOX 360





Four viewpoints on AI

	Humanly	**Rationally**
Thinking	**Think like humans** *Cognitive modelling approach*	**Think rationally** “Laws of thought” / logic approach
Acting	**Act like humans** *Turing Test approach*	**Act rationally** *Rational-agent approach*





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Mentimeter 6

Which Viewpoint defines AI the best?



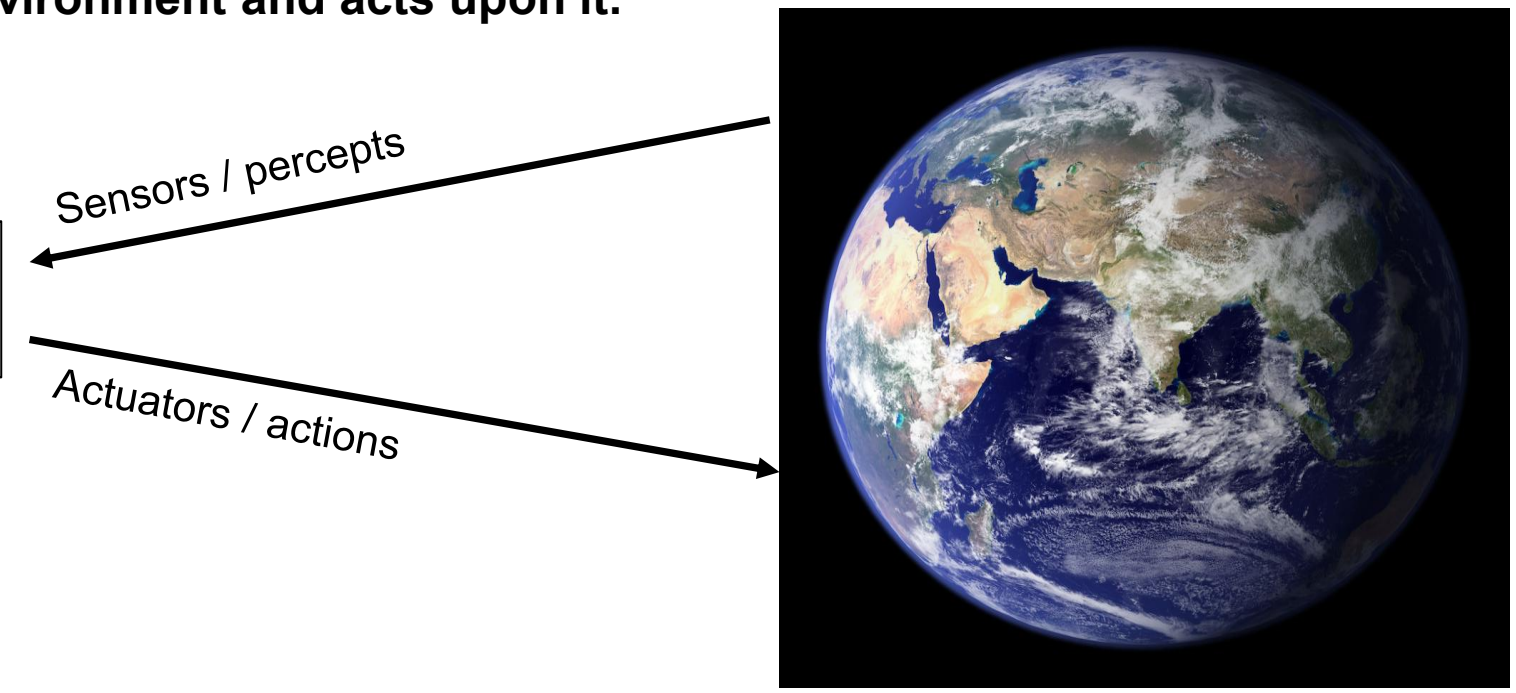
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Act rationally: intelligent agents

- *"AI: A Modern Approach"* covers all four AI viewpoints, but primarily adopts the **rational-agent approach**
- Rational behavior (doing the right thing given the available information): choosing the action expected to maximize the performance measure.
- Perfect rationality may be impossible because of computational and imperfect information limitations.
- **An agent perceives its environment and acts upon it.**

Agent function:

$$f : \mathcal{P}^* \rightarrow \mathcal{A}$$



Assembly robot arm



- What are the performance measures?
  - Percentage of correctly assembled objects, speed, damage/safety, energy/cost
- What are the sensor inputs?
  - Camera images, joint positions, force-torque readings
- What are the actions?
  - Arm movements, gripper commands
- What is the environment?
  - Conveyer belt with pieces, *partially observable*, *stochastic*, sequential, dynamic, continuous, single-agent

Environment properties

Property	Meaning
Fully vs partially observable	Does the agent have access to all relevant state information?
Deterministic vs stochastic	Does an action uniquely determine the next state?
Episodic vs sequential	Do previous actions affect future decisions?
Static vs dynamic	Can the environment change while the agent is deciding?
Discrete vs continuous	Are states/actions/time discrete or continuous?
Single vs multi-agent	Are there other agents whose actions matter?
Cooperative vs competitive	Are the agents' objectives aligned or opposed?

Mentimeter 7-10

- Think, Pair, Share
- For an autonomous car **and** for a chess engine
  - What are the performance measures?
  - What are the inputs?
  - What are the outputs?
  - What are the environment properties?
    - Observability (Full/Partial)
    - Deterministic vs Stochastic
    - Episodic vs sequential
    - Static vs Dynamic
    - Discrete vs Continuous
    - Single agent vs Multi agent
    - Cooperative vs competitive



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AI is the ability of an artificial machine to

- Solve problems
  - search, adversarial search
- Reason, plan, make decisions
  - logic, inference, planning algorithms, reinforcement learning
- Learn
  - machine learning, neural nets
- Communicate and perceive
  - computer vision, perception, speech recognition and synthesis, natural language processing
- Move and manipulate objects
  - robotics

